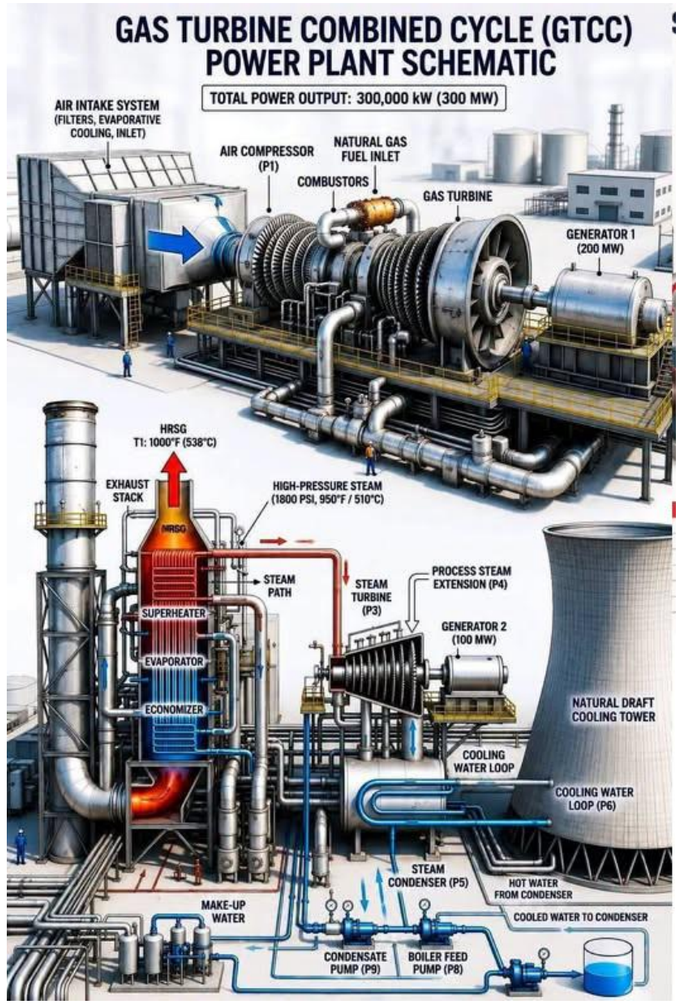
STEAM / GAS TURBINES MAINTENANCE FRAGNET

- **Initial Inspection and Preparation**
  - **Pre-Maintenance**
    - Verify the turbine's operational history, including any past failures or deviations.
    - Gather and review technical documentation: operational manuals, OEM maintenance guidelines, and historical maintenance logs.
    - Confirm the availability of necessary tools, spare parts, and qualified personnel.
  - **ISOLATE / LOTO**
    - Shut down the turbine according to the manufacturer's guidelines.
    - Verify that all energy sources, including steam, electricity, and hydraulic systems, are safely isolated.
    - Ensure Lockout/Tagout (LOTO) protocols are strictly followed.
- **External Inspection**
  - **Casing and Insulation**
    - Inspect the turbine casing for cracks, leaks, or deformation.
    - Check for wear and tear on insulation materials and ensure proper sealing.
    - Assess bolt tension and flange alignment to prevent steam leakage.
  - **Lubrication System**
    - Analyze oil samples for viscosity, contamination, and degradation.
    - Inspect oil filters and replace them if necessary.
    - Verify the functionality of pumps, coolers, and heaters within the lubrication system.
- **Rotor & Blades**
  - **Rotor Inspection**
    - Measure rotor runout and compare it against OEM tolerances.
    - Check for signs of thermal distortion, pitting, or erosion on the shaft.
    - Conduct non-destructive testing (NDT) such as ultrasonic or dye penetrant testing to detect cracks.
  - **Blade Inspection**
    - Inspect blades for corrosion, cracking, or deposits that may affect performance.
    - Verify the integrity of blade attachments and ensure no looseness.
    - Assess the leading and trailing edges for wear, erosion, or impact damage.
- **Bearing & Seals**
  - **Bearing Maintenance**
    - Measure bearing clearances and check alignment.
    - Replace worn or damaged bearings.
    - Verify proper lubrication flow and pressure.
  - **Seal Integrity**
    - Inspect labyrinth seals, mechanical seals, or carbon seals for wear or leakage.
    - Replace seals that exhibit excessive clearance or deformation.
- **Steam/Gas Path Components**
  - **Nozzles and Diaphragms**
    - Examine nozzle vanes for scaling, erosion, or fouling.
    - Inspect diaphragms for cracks or misalignment and repair as necessary.
  - **Control Valves**
    - Test and calibrate control valves for accurate operation.
    - Inspect valve seats, stems, and bushings for wear or damage.
- **Auxiliary Systems**
  - **Steam Piping**
    - Conduct a thermal expansion check and inspect pipe supports for integrity.
    - Test safety valves for proper set points and response.
  - **Condensers and Cooling Systems**
    - Clean condenser tubes to remove fouling and ensure efficient heat transfer.
    - Check cooling water flow and inspect for leaks or blockages.
- **Dynamic Testing**
  - **Vibration Analysis**
    - Conduct a vibration analysis to detect imbalances, misalignments, or mechanical looseness.
    - Compare vibration levels to baseline data and OEM limits.
  - **Performance Testing**
    - Verify steam flow rates, pressure ratios, and exhaust conditions.
    - Conduct efficiency testing to identify potential performance degradation.
- **Post-Maintenance Procedures**
  - **Reassembly and Alignment**
    - Ensure all components are reassembled according to OEM specifications.
    - Conduct precision alignment of the rotor to prevent stress on bearings.
  - **Startup Protocol**
    - Gradually bring the turbine up to operational speed, monitoring for unusual noises or vibrations.
    - Verify system parameters, including temperature, pressure, and flow rates, during startup.
- **Documentation and Reporting**
  - Record all inspection findings, maintenance actions, and test results in the turbine's logbook.
  - Provide detailed reports to consultants or stakeholders, highlighting any concerns or recommendations.
  - Update the preventive maintenance schedule based on findings.

	Existing LP/MP Turbine Dismantling							
	300	Disconnecting of all Lube oil Lines connected to turbine-	12	100%	15-Feb-26 20:00	16-Feb-26 08:00	15-Feb-26 09:00	18-Feb-26 00:00
	301	Removal of all instrumentations such as RTD's, vibraton	8	100%	15-Feb-26 20:00	16-Feb-26 04:00	16-Feb-26 11:00	17-Feb-26 18:00
	302	Removal of main steam I/O Lines-Cold work	24	100%	15-Feb-26 20:00	16-Feb-26 20:00	15-Feb-26 12:00	18-Feb-26 12:00
	303	Removal of main steam I/O Lines-Hot work	24	100%	18-Feb-26 08:00	19-Feb-26 08:00	15-Feb-26 08:00	19-Feb-26 11:00
	304	Removal of existing GSC and its associated piping	10	100%	20-Feb-26 08:00	20-Feb-26 18:00	17-Feb-26 09:00	17-Feb-26 22:00
	305	Removal of existing Gland sealing station and its associ	10	100%	20-Feb-26 18:00	21-Feb-26 04:00	18-Feb-26 00:00	22-Feb-26 08:00
	306	Dismantle old Control oil system & piping, and other oil	12	100%	18-Feb-26 08:00	18-Feb-26 20:00	18-Feb-26 00:00	19-Feb-26 11:00
	307	Locking the turbine at front and rear ends	0	100%	21-Feb-26 04:00	21-Feb-26 08:00	17-Feb-26 00:00	17-Feb-26 18:00
	308	Disassembly of coupling guards between Turbine to con	8	100%	15-Feb-26 20:00	16-Feb-26 04:00	18-Feb-26 04:00	19-Feb-26 19:00
	309	Disassembly of couplings between HP Turbine to LP cor	8	100%	16-Feb-26 04:00	16-Feb-26 12:00	16-Feb-26 23:00	17-Feb-26 17:00
	310	De-bolt and disconnect exhaust spool and Bellow	24	100%	18-Feb-26 08:00	19-Feb-26 08:00	17-Feb-26 09:00	19-Feb-26 18:00
	311	Removal Of Turbine holding on Bolts , Lifting of Turbine	6	100%	21-Feb-26 08:00	21-Feb-26 14:00	20-Feb-26 00:00	20-Feb-26 22:00
	312	Dismantle old LP turbine from its position.	8	100%	21-Feb-26 14:00	21-Feb-26 22:00	20-Feb-26 00:00	20-Feb-26 18:00
	313	Dismantle of exhaust spool and Bellow	12	100%	21-Feb-26 22:00	22-Feb-26 10:00	20-Feb-26 00:00	20-Feb-26 17:00
	314	Dismantle old HP turbine from its position.	8	100%	21-Feb-26 22:00	22-Feb-26 06:00	20-Feb-26 09:00	20-Feb-26 20:00
	315	Remove existing turbine pedastals -HPT+LPT	8	100%	22-Feb-26 06:00	22-Feb-26 14:00	21-Feb-26 13:00	21-Feb-26 22:00
	Base Frame modifications Welding & fabrication & Turbine erection							
	317	Base frame cleaning - Cutting, grinding new Base frame	12	100%	22-Feb-26 14:00	23-Feb-26 02:00	20-Feb-26 16:00	22-Feb-26 20:00
	318	Bellow with transition spool erection to be done	24	100%	22-Feb-26 10:00	23-Feb-26 10:00	27-Feb-26 00:00	28-Feb-26 00:00
	319	Marking reference dimension	12	100%	23-Feb-26 02:00	23-Feb-26 14:00	19-Feb-26 00:00	22-Feb-26 00:00
	320	Position sole plate	12	100%	23-Feb-26 06:00	23-Feb-26 18:00	22-Feb-26 00:00	28-Feb-26 00:00
	321	Blue matching of sole plate with Base frame	12	100%	23-Feb-26 18:00	24-Feb-26 06:00	22-Feb-26 12:00	28-Feb-26 00:00
	322	Locking sole plate and Erection of New Syngas turbine	8	100%	24-Feb-26 06:00	24-Feb-26 14:00	27-Feb-26 00:00	04-Mar-26 00:00
	323	Vertical elevation adjustment and rough alignment	8	90%	24-Feb-26 14:00	24-Feb-26 22:00	04-Mar-26 00:00	
	334	Final alignment of Turbine and Sole plate	8	80%	01-Mar-26 20:00	02-Mar-26 04:00	04-Mar-26 00:00	
	Perform the welds in the specific sequence indicated on the released drawing 0-50740-E42116A-00							
	325	Weld part 1 beam with existing base frame	10	100%	24-Feb-26 22:00	25-Feb-26 08:00	21-Feb-26 00:00	22-Feb-26 00:00
	326	Weld part 2 beam with existing base frame	20	100%	25-Feb-26 08:00	26-Feb-26 04:00	19-Feb-26 00:00	19-Feb-26 23:00
	New Syngas Turbine Alignment activites							
	336	Centre Line marking of Turbine	12	100%	02-Mar-26 04:00	02-Mar-26 16:00	21-Feb-26 00:00	22-Feb-26 00:00
	337	Bearings inspection	10	50%	02-Mar-26 16:00	03-Mar-26 02:00	04-Mar-26 00:00	
	338	Tubine centering to be checked and if necessary adjuste	8	50%	03-Mar-26 02:00	03-Mar-26 10:00	04-Mar-26 00:00	
	339	Final assembly of bearings	10	50%	03-Mar-26 10:00	03-Mar-26 20:00	04-Mar-26 00:00	
	340	Alignment between turbine to compressors to be checke	48	50%	03-Mar-26 20:00	05-Mar-26 20:00	04-Mar-26 00:00	
	341	MS line tightening studs new /old if old the same should	4	50%	05-Mar-26 20:00	06-Mar-26 00:00	04-Mar-26 00:00	
	342	connect exhaust spool to turbine	12	50%	05-Mar-26 20:00	06-Mar-26 08:00	04-Mar-26 00:00	
	343	Re check the turbine centering after steam pipeline conr	4	50%	06-Mar-26 00:00	06-Mar-26 04:00	04-Mar-26 00:00	
	344	Checking of all holding down bolts and maintained the r	10	50%	06-Mar-26 04:00	06-Mar-26 14:00	04-Mar-26 00:00	
	345	Coupling & Coupling Guard fixing and Modification if rec	12	50%	06-Mar-26 14:00	07-Mar-26 02:00	04-Mar-26 00:00	
	346	Install steam blowing device on ESV	4	50%	06-Mar-26 14:00	06-Mar-26 18:00	04-Mar-26 00:00	
	347	Assembly of all servomotors, Control Valve & Barring ge	10	50%	06-Mar-26 14:00	07-Mar-26 00:00	04-Mar-26 00:00	
	348	Assembly of ESV after steam blowing	4	50%	12-Mar-26 08:00	12-Mar-26 12:00	04-Mar-26 00:00	
	349	Final boxup of all the bearings with RTD's	4	50%	07-Mar-26 00:00	07-Mar-26 04:00	04-Mar-26 00:00	
	350	Assembly of all the instruments such as vibration probes	10	0%	07-Mar-26 04:00	07-Mar-26 14:00		
	351	Electrical connection of the barring gear motor (before c	8	0%	07-Mar-26 14:00	07-Mar-26 22:00		
	New Syngas Turbine Alignment activites							
	336	Centre Line marking of Turbine	12	100%	02-Mar-26 04:00	02-Mar-26 16:00	21-Feb-26 00:00	22-Feb-26 00:00
	337	Bearings inspection	10	50%	02-Mar-26 16:00	03-Mar-26 02:00	04-Mar-26 00:00	
	338	Tubine centering to be checked and if necessary adjuste	8	50%	03-Mar-26 02:00	03-Mar-26 10:00	04-Mar-26 00:00	
	339	Final assembly of bearings	10	50%	03-Mar-26 10:00	03-Mar-26 20:00	04-Mar-26 00:00	
	340	Alignment between turbine to compressors to be checke	48	50%	03-Mar-26 20:00	05-Mar-26 20:00	04-Mar-26 00:00	
	341	MS line tightening studs new /old if old the same should	4	50%	05-Mar-26 20:00	06-Mar-26 00:00	04-Mar-26 00:00	
	342	connect exhaust spool to turbine	12	50%	05-Mar-26 20:00	06-Mar-26 08:00	04-Mar-26 00:00	
	343	Re check the turbine centering after steam pipeline conr	4	50%	06-Mar-26 00:00	06-Mar-26 04:00	04-Mar-26 00:00	
	344	Checking of all holding down bolts and maintained the r	10	50%	06-Mar-26 04:00	06-Mar-26 14:00	04-Mar-26 00:00	
	345	Coupling & Coupling Guard fixing and Modification if rec	12	50%	06-Mar-26 14:00	07-Mar-26 02:00	04-Mar-26 00:00	
	346	Install steam blowing device on ESV	4	50%	06-Mar-26 14:00	06-Mar-26 18:00	04-Mar-26 00:00	



Drw Purpose Code :- R01- Educational	**Statutory Note:** This document is prepared by rPlan and is the sole proprietory of the company.Its purpose is educational and no references within this document are for Engineering or Design corelation with the applicable codes. For any clarifications beyond the intended purpose viewers can always correspond to rPlan via its company website.This drw is not for duplication without company permission!	Description: Typical FRAGNET Turbine Maintenance	
Status of Approval :- 01			
Dimentions as Mentioned NA	Drw electronic code		
Drawing Scale :- NTS	rPlan AT 39, MERMAID 1 CHS LTD., SEC-11, CBD-BELAPUR, NAVI MUMBAI - 400614	Drawn By: DFT-01	Drawn Date 20260729
Engg Code :- 02 / Backup Info Code :- 04	Legal Owner / QR CODE	Checked Approved by rPlan-01	Checked Approved Date 20260802
Angle of Projection NA / Sheet Size A3	NA	Drawing Number As Barcoaded / Sheet 1 of 1 / Rev 00	